Time Series Selection and Splitting

A) Dataset Selection
`TSPredIT` Series

B) Data Splitting
Train Set: $\{x_1, x_2, ..., x_{t-1}, x_t\}$
Test Set: $\{x_{t+1}, x_{t+2}, ..., x_{T-1}, x_T\}$

C. *Machine Learning*

● : ForeSight
● : ReW / CReW
● : Others

Model Training and Tuning

C1) Nonstationarity Handling
• Differencing, Smoothing, and Transformation Techniques

C2) Data Normalization
• Global Min-Max, Sliding Window Min-Max, Other Normalization Techniques

C3) Time Series Windowing
• Window Size

C4) Temporal Encoding
• Positional, Fourier, RFF, MSRFF, ConvSPE, TRep, TS2Vec
• Embedding Dimension

C5) Binary Encoding
• Distributed and Gaussian Thermometers
• Number of Bits

C6) Training and Hyperparameter Optimization
• General (Normalization, Window Size) and Model-Specific (ML, WNNs)

D. ARIMA

D1) Order-*d* Differencing
$\nabla^d x_i = x_i - x_{i-1}$

D2) AR(ρ) and MA(*q*) Model Optimization

E. Chronos

E1) Data Normalization
Standardization

E2) Data Augmentation
• TSMixup
• KernelSynth

E3) Tokenization

E4) Context Building

E5) Inference Strategies
• Zero-Shot
• Fine-Tuning

Testing and Results

F) Forecasting and Post-Processing
• Rolling Origin and Steps Ahead
• Naïve Forecast (WNNs)
• Inverse Transformations

G) Performance Metrics
• sMAPE and Execution Time
• Estimated Energy Consumption

H) Model Evaluation
• Statistical Significance, Pareto Frontier, Model Ranking
• Qualitative Analysis